Assignment 3

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1. **Interpretation of Relationships Between Variables**

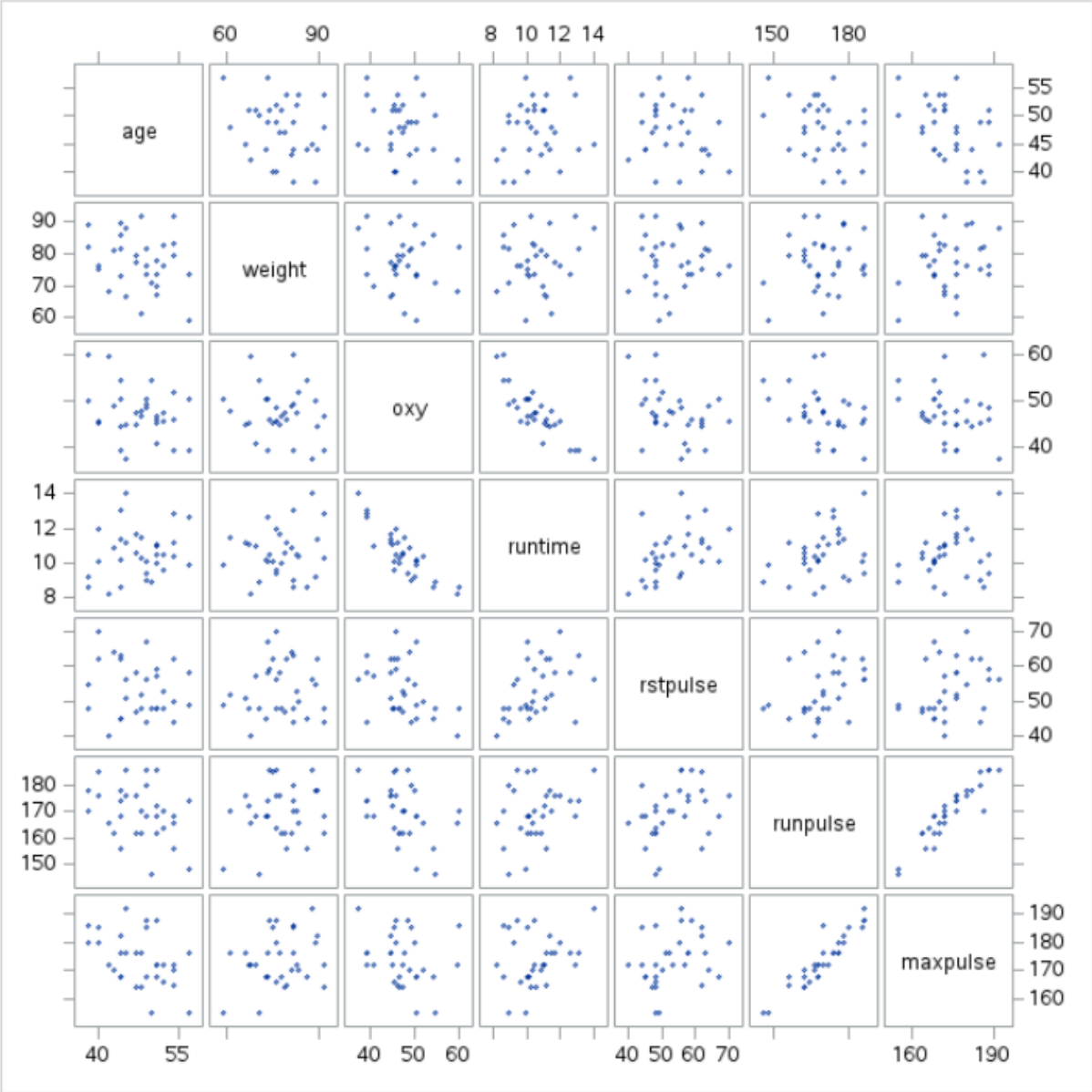

Figure 1: Figure X. Full Scatterplot Matrix of All Variables in the Fitness Dataset

This section describes the relationships between the physiological and performance variables in the Fitness dataset, using both scatter plots and correlation coefficients. Each interpretation is linked to a specific figure that should be inserted in the report. Before analyzing each pair of variables, I first examined the full scatterplot matrix , which provides a visual overview of all relationships in the dataset. This matrix helps identify general patterns, such as linear trends, clusters, or potential outliers. The detailed interpretations below refer to the individual scatterplots contained in this matrix.

- **oxy and runtime**

Correlation: –0.86219 (very strong negative) This is the strongest relationship in the dataset. The negative sign means that higher oxygen consumption is associated with lower running time. In other words, people who can take in more oxygen during exercise tend to run faster. This makes perfect sense because oxygen consumption is a direct indicator of aerobic fitness. The strength of the correlation shows that this relationship is extremely consistent across individuals.

- **oxy and rstpulse**

Correlation: –0.39936 (moderate negative) People with higher oxygen consumption tend to have lower resting heart rates. This is a classic sign of cardiovascular fitness: a stronger heart pumps more efficiently and doesn't need to beat as often at rest. The relationship is not extremely strong, but it is clearly present.

- **oxy and runpulse**

Correlation: –0.39797 (moderate negative) This relationship is similar to the previous one. Individuals with higher oxygen consumption tend to have lower heart rates during the run. This suggests that their cardiovascular system is more efficient even under physical stress. The moderate strength indicates that this pattern is noticeable but not perfect.

- **oxy and maxpulse**

Correlation: –0.23674 (weak negative) There is a slight tendency for people with higher oxygen consumption to have lower maximum pulse values. However, the relationship is weak, meaning it is not reliable. Maximum pulse is influenced by age and genetics more than fitness level, which explains why the correlation is small.

- **oxy and weight**

Correlation: –0.16275 (weak negative) Weight has almost no relationship with oxygen consumption in this dataset. This means that heavier individuals are not necessarily less fit, and lighter individuals are not automatically more fit. Other factors like training level and lifestyle play a much bigger role.

- **oxy and age**

Correlation: –0.30459 (weak to moderate negative) Older individuals tend to have slightly lower oxygen consumption. This is expected because aerobic capacity naturally declines with age. However, the relationship is not very strong, which suggests that some older participants may still be quite fit.

- **runtime and rstpulse**

Correlation: 0.45038 (moderate positive) People who take longer to complete the run tend to have higher resting heart rates. This indicates lower baseline fitness. The moderate strength shows that this relationship is meaningful and consistent.

- **runtime and runpulse**

Correlation: 0.31365 (weak to moderate positive) Slower runners tend to have higher heart rates during the run. This makes sense because less fit individuals need to work harder to maintain the same pace. The relationship is weaker than runtime–rstpulse, but still noticeable.

- **runtime and maxpulse**

Correlation: 0.22610 (weak positive) There is a slight tendency for slower runners to reach higher maximum pulse values. However, the relationship is weak, meaning it is not a reliable predictor. Maximum pulse is influenced by age and individual physiology.

- **runtime and weight**

Correlation: 0.14351 (weak positive) Weight has almost no effect on running time in this sample. Some heavier individuals may still run fast, and some lighter individuals may run slow. The relationship is too weak to draw conclusions.

- **runtime and age**

Correlation: 0.18875 (weak positive) Older individuals tend to run slightly slower, but the relationship is weak. Age alone does not explain much of the variation in running time.

- **rstpulse and runpulse**

Correlation: 0.35246 (weak to moderate positive) People with higher resting heart rates also tend to have higher heart rates during the run. This is logical because resting heart rate reflects baseline cardiovascular condition. The relationship is noticeable but not extremely strong.

- **rstpulse and maxpulse**

Correlation: 0.30512 (weak to moderate positive) Individuals with higher resting pulse tend to have slightly higher maximum pulse. This suggests that some people naturally have higher heart rate ranges. The relationship is not very strong, but it is present.

- **rstpulse and weight**

Correlation: 0.04397 (very weak positive) Weight has almost no relationship with resting heart rate. This means that heavier individuals do not necessarily have higher resting pulse.

- **rstpulse and age**

Correlation: –0.16410 (weak negative) Older individuals tend to have slightly lower resting pulse, but the relationship is very weak. This is not a meaningful predictor.

- **runpulse and maxpulse**

Correlation: 0.92975 (very strong positive) This is the second strongest relationship in the dataset. People with high running pulse almost always have high maximum pulse. These two variables are extremely similar and almost interchangeable. This makes sense because both measure heart rate under physical stress.

- **runpulse and weight**

Correlation: 0.18152 (weak positive) Weight has a very small effect on running pulse. The relationship is too weak to be meaningful.

- **runpulse and age**

Correlation: –0.33787 (weak to moderate negative) Older individuals tend to have slightly lower heart rates during the run. This is consistent with the natural decline in heart rate capacity with age.

- **maxpulse and weight**

Correlation: 0.24938 (weak positive) Heavier individuals tend to have slightly higher maximum pulse, but the relationship is weak. This is not a reliable pattern.

- **maxpulse and age**

Correlation: -0.43292 (moderate negative) Older individuals have lower maximum heart rates. This is a well-known physiological fact and one of the clearer relationships in the dataset.

- **weight and age**

Correlation: –0.23354 (weak negative) Older individuals in this sample tend to weigh slightly less, but the relationship is weak. This could be due to lifestyle differences or sample characteristics.

2. Limitations of the Correlation Coefficient for the Fitness Data

Pearson Correlation Coefficients, N = 31
Prob > |r| under H0: Rho=0

	age	weight	oxy	runtime	rstpulse	runpulse	maxpulse
age	1.00000	-0.23354 0.2061	-0.30459 0.0957	0.18875 0.3092	-0.16410 0.3777	-0.33787 0.0630	-0.43292 0.0150
weight	-0.23354 0.2061	1.00000	-0.16275 0.3817	0.14351 0.4412	0.04397 0.8143	0.18152 0.3284	0.24938 0.1761
oxy	-0.30459 0.0957	-0.16275 0.3817	1.00000	-0.86219 <.0001	-0.39936 0.0260	-0.39797 0.0266	-0.23674 0.1997
runtime	0.18875 0.3092	0.14351 0.4412	-0.86219 <.0001	1.00000	0.45038 0.0110	0.31365 0.0858	0.22610 0.2213
rstpulse	-0.16410 0.3777	0.04397 0.8143	-0.39936 0.0260	0.45038 0.0110	1.00000	0.35246 0.0518	0.30512 0.0951
runpulse	-0.33787 0.0630	0.18152 0.3284	-0.39797 0.0266	0.31365 0.0858	0.35246 0.0518	1.00000	0.92975 <.0001
maxpulse	-0.43292 0.0150	0.24938 0.1761	-0.23674 0.1997	0.22610 0.2213	0.30512 0.0951	0.92975 <.0001	1.00000

Figure 2: Correlation Table

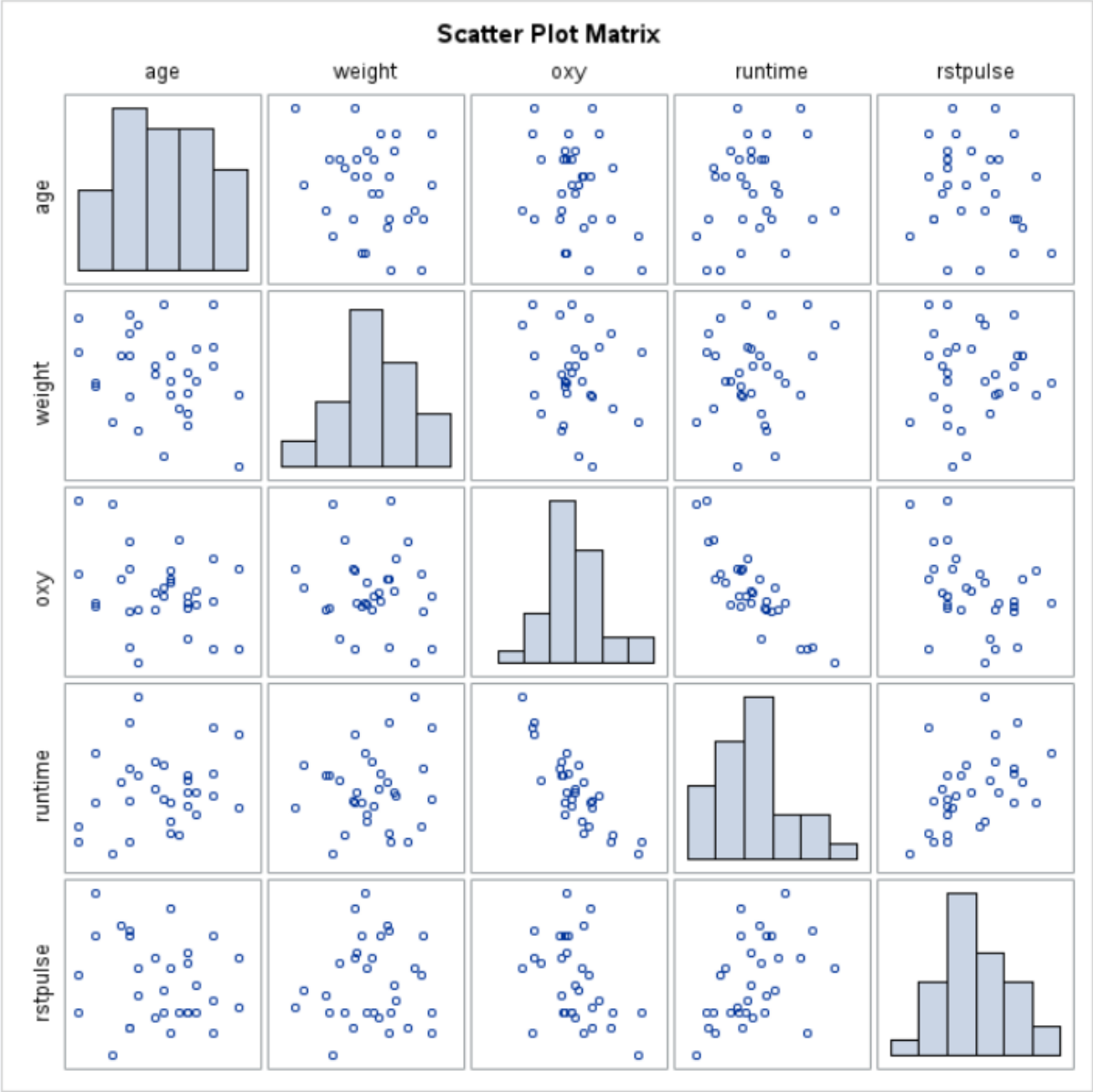

Figure 3: Histogram Analysis

Even though the correlation coefficient is useful for understanding relationships between variables in the Fitness dataset, it has several important limitations. These limitations affect how we interpret the results and how much we can trust the conclusions. Below are the main limitations, explained in the context of this dataset.

a. Correlation does not imply causation

A strong correlation between two variables does not mean that one causes the other. For example:

- oxy and runtime have a very strong negative correlation (–0.86), but this does not *prove* that increasing oxygen consumption will automatically make someone run faster. Other factors like training, genetics, or technique could influence both variables.

In the Fitness data, correlations show associations, not direct cause-and-effect relationships.

b. Pearson correlation only measures linear relationships

The correlation coefficient assumes that the relationship between two variables is linear. If the relationship is curved or more complex, Pearson's r will not capture it.

For example:

- weight and runtime show a weak correlation (0.14), but this does not mean weight has no effect. It could be that the relationship is nonlinear (e.g., both very low and very high weights affect performance differently).

Because Pearson only detects straight-line patterns, some relationships may be underestimated.

c. Sensitivity to outliers

Pearson's r can be strongly affected by extreme values. Even one unusual observation can change the correlation dramatically.

In the Fitness dataset:

- If one participant had an extremely high pulse or an unusually slow runtime, it could distort the correlation values. Even though the histograms show no major outliers, the sample is small (n = 31), so one extreme case would have a big impact.

d. Correlation does not control for other variables

Pearson's r only looks at two variables at a time. It does not consider the influence of other factors.

For example:

- age affects maxpulse
- age also affects oxy
- oxy affects runtime

But the correlation between oxy and runtime does not adjust for age. This means some correlations may be inflated or reduced because of hidden variables.

A more advanced method like multiple regression would be needed to control for these effects.

e. Small sample size (n = 31)

With only 31 participants, the correlations may not be stable or generalizable. Small samples increase the chance that:

- moderate correlations appear stronger or weaker than they really are
- weak correlations may not reach statistical significance
- random variation affects the results

In a larger dataset, some of the weak correlations (like weight vs oxy) might disappear completely or become stronger.

f. Correlation does not distinguish between meaningful and meaningless relationships

Some correlations in the dataset are statistically weak but still physiologically meaningful, while others are statistically moderate but not very useful.

For example:

- age vs maxpulse (–0.43) is meaningful because it reflects a known biological pattern
- runtime vs weight (0.14) is weak and not very informative

Pearson's r treats all relationships the same way, even though their real-world importance differs.

g. Correlation ignores measurement error

If any variable is measured inaccurately (e.g., pulse taken incorrectly), the correlation will be affected.

In the Fitness dataset:

- heart rate measurements can vary depending on technique
- oxygen consumption can fluctuate depending on effort
- runtime can be influenced by pacing or motivation

Pearson's r assumes perfect measurement, which is rarely true in real data.